

OKOLIE CHUKWUKA

AI & Software Engineer | Machine Learning | LLM Systems | Chatbot | Voice AI

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Summary

Machine Learning Engineer and AI Software Engineer focused on scalable systems, deployment, and operations. Expertise in LLMs, Voice AI, Computer Vision, and multi-agent workflows. Built production-grade RAG, chatbots, and speech systems (STT/TTS) across healthcare, fintech, and govtech. Strong MLOps (Docker, MLflow, GitHub Actions), with hands-on AWS deployments; lead ML teams to deliver reliable, user-facing AI products.

Experience

AI & Software Engineer

MBL HighTech • Dec 2024 – Present

- Built AI voice-based telephony systems; increased lead conversion by 20% via prompt/call-flow optimization
- Designed LLM chatbots for daily user engagement; cut response latency by 50% through retrieval and caching
- Architected LangGraph + FastAPI + Postgres pipelines for multi-agent workflows and CRM integration
- Implemented ML CI/CD (GitHub Actions) and Docker for repeatable releases; improved deployment time by 30%

Machine Learning Engineer

NewWave Telecom & Technologies Inc (USA) • Aug 2023 – Jun 2024

US federal health IT contractor (data & AI for large agencies)

- Built an AI-powered RFP engine (CrewAI + LLMs) to extract/synthesize content; cut drafting time from ~30 days to under 4 hours
- Generated structured text, tables, and visuals (graphs/flowcharts); increased proposal throughput by 20%
- AWS usage: S3 for corp docs/artifacts; EC2 for orchestration runners

Lead ML Engineer

Glodaris Technology (Abuja) • Jan 2022 – Mar 2023

Abuja-based technology company (computer vision & analytics)

- Led 3+ engineers to develop Face Recognition System (97.3% accuracy) for Nigerian border validation
- Designed real-time model monitoring microservices (Prometheus/Grafana) and trained officers on ML systems and Python

Machine Learning Engineer (Contract)

Indicina • Jan 2021 – Jul 2021

Website: indicina.co

Fintech credit decisioning platform (digital lending)

- Built core components of DECIDE AI credit platform; improved risk scoring AUC by 30%
- Developed predictive risk models and ML pipelines; partnered with product/risk, reducing manual review by 5%
- Tooling: scikit-learn, XGBoost, Pandas; APIs with FastAPI

AI Engineer (Contract)

AirSmat • Jan 2020 – Nov 2020

Website: airsmat.com

Drone analytics for precision agriculture

- Delivered drone-imagery pipelines for crop yield and disease detection (object detection)
- Data pipelines: S3 object storage for geospatial tiles; batch processing on EC2



Project Highlights

Founder — NaijaLingo

In Development

- Speech AI platform offering TTS + STT for Yoruba, Igbo, Hausa, and Pidgin English
- Authentic voice synthesis with controllable emotion, tone, and accent for inclusive communication



Freelance Project Highlights (2023–Present)

36+ global clients • 50+ successful projects • 4.9★ rating

Text Analysis App

NLP web app with TF-IDF, LDA, BERTopic, and Sentence-BERT embeddings

STA Automation AI

AI-powered test automation with Llama 3 and Robot Framework

Theoriq Docs Assistant

RAG-based agent using Embedchain and Flask

LLM based Text and Audio data Model Training and Fine-tuning

Trained a variety of models for text and audio data using LLMs, TTS and STT models and fine-tuned them for specific tasks.

Investroj.cz

Automated property data collection, analysis, and visualization (sreality.cz).

DAO Voting Agent

FastAPI + Web3.py + LangChain for conversational blockchain voting

Naija Messenger

Golang app automating WhatsApp birthday wishes with AI



Education

B.Sc. Physics

University of Benin



Certifications

- Machine Learning with Python: Zero to GBMs — Jovian.ai
- Deep Learning with PyTorch: Zero to GANs
- AI for Medical Diagnosis — Coursera



Soft Skills

Leadership • Communication • Mentorship • Stakeholder Management • Problem Solving • Collaboration



Core Tech Stack

Artificial Intelligence • Machine Learning • Data Science • Python • Golang • Transformers • LLMs • LangChain • LangGraph • OpenAI API • CrewAI • FastAPI • Flask • PostgreSQL • Redis • Docker • CI/CD • MLflow • Airflow • DVC • Feature Store (Feast) • PyTorch • TensorFlow • scikit-learn • XGBoost • RAG • Whisper (STT) • TTS • Speech AI • Voice Cloning • MLOps • AWS • S3 • EC2 • Lambda • RDS • SageMaker • EKS • CloudWatch • Neo4j